Future Epics & Post-Hackathon Rollout

Engineering Specifications for Horizon 1 (Field Expansion), Horizon 2 (Community Scaling) & Milestone 6 National TCC Pilot (2026 - 2027)

1. Post-Hackathon Strategic Roadmap Architecture

To satisfy the FNBAOTY mentor mandate of **avoiding scope overload while building for scalable national impact**, Umkhuseli isolates advanced engineering capabilities into two phased horizons. Foundational components for these epics already exist within the repository (e.g. SafeJourneyManager.kt, audioSentryService.ts, CarrierDetectionManager.kt, and WifiDirectShareManager.kt). By maintaining clean architectural boundaries, the 5 MVP Core features remain lightweight, ultra-reliable, and strictly offline-first, while providing a clear blueprint for institutional expansion.

PRIVACY & RELIABILITY GUARDRAILS

Architectural Decoupling Principle: Future Epics operate as modular extensions plugged into the existing AppContainer. They are gated behind user consent and strictly adhere to the POPIA principle of minimal data collection. No future epic compromises the 0-ms instant camouflage scrub or local-first evidence vault.

2. Horizon 1: Immediate Field Expansion (Phase 2 — Q1 2027)

Horizon 1 focuses on high-risk transit environments and automated distress recognition:

Epic ID & Name	Technical Architecture & Implementation	South African Context & Survivor Value
[EPIC-01] Safe Journey E-Hailing Corridor Sentinel	Implemented via SafeJourneyManager.kt. Leverages fused location provider to track transit corridors for Bolt, Uber, and minibus taxi rides. Triggers dead-man's periodic check-in; escalates to silent ICE dispatch if off-route by >250m.	Protects commuters and students traveling late through high-risk urban routes (e.g. Johannesburg CBD, Durban Warwick Triangle) against route diversion or abduction.
[EPIC-02] Audio Sentry Ambient Acoustic Classifier	Implemented via audioSentryService.ts. On-device lightweight acoustic classifier analyzing 15-second circular RAM buffer for decibel spikes and human screaming frequencies (1 kHz - 4 kHz). Never streams audio to cloud.	Automates emergency dispatch when a survivor is physically restrained or cannot reach their handset to enter unlock codes or trigger panic gestures.
[EPIC-03] Direct Carrier VAS & Reverse-Billing	Implemented via CarrierDetectionManager.kt. Integrates telco VAS APIs with Vodacom, MTN, Telkom, and Cell C to facilitate zero-rated USSD reverse-billing (*120*7867#) and automatic data token replenishment.	Guarantees emergency connectivity for township residents with R0.00 airtime balance, eliminating poverty-induced communication blackouts.

3. Horizon 2: Community & Ecosystem Scaling (Phase 3 — Q2/Q3 2027)

Horizon 2 introduces decentralized cryptographic trust and community-level rapid response networks:

Epic ID & Name	Technical Architecture & Implementation	South African Context & Survivor Value

[EPIC-04] OPRF PSI Private Contact Discovery	2-Hash Elliptic Curve Diffie-Hellman Oblivious PRF over Curve25519. Allows survivors to verify if trusted contacts or advocates use Umkhuseli without uploading address books or exposing phone numbers to rainbow table attacks.	Prevents abusive partners, stalkers, or corrupt telecommunication operators from scanning subscriber databases to identify who has installed safety applications.
[EPIC-05] 'Touched by an Angel' Citizen First-Responders	Implemented via GuardianResponderSection.kt. Radial dispatch algorithm (500m - 2km). Requires 3-tier KYC: DHA National ID verification, biometric liveness check, and SAPS Police Clearance Certificate.	Overcomes SAPS 45+ minute rural response delays by mobilizing vetted local community aunties, church leaders, and CPF members within walking distance.
[EPIC-06] P2P Wi-Fi Direct Covert Mesh Share	Implemented via WifiDirectShareManager.kt and CovertNearbyShareSection.kt. Localized peer-to-peer APK and offline TCC database replication via Wi-Fi Direct / BLE without cellular or internet connectivity.	Enables Community Health Workers (CHWs) to safely distribute the application to women in deep rural areas or informal settlements lacking mobile network coverage.
[EPIC-07] Ward-Level Pseudonymous Safety Circles	PostGIS ward-level polygon geofencing coupled with monthly rotating cryptographic pseudonyms and zero-knowledge location proofs.	Provides localized alert boards where survivors can share anonymous safety warnings (e.g. violent perpetrators, dark taxi corridors) without being identified.

4. Milestone 6: National Rollout & Operational Partnerships (2027)

The national rollout will scale Umkhuseli across institutional public health, policing, and justice frameworks:

- **5-Facility TCC Pilot:** Live field trials embedded in high-volume Thuthuzela Care Centres: Chris Hani Baragwanath (Soweto), Helen Joseph (JHB), Kalafong (Atteridgeville), Karl Bremer (Cape Town), and Prince Mshiyeni (Umlazi).
- **11-Language Localization:** Native localization across all 11 official South African languages (isiZulu, isiXhosa, Afrikaans, Sepedi, Setswana, Sesotho, Xitsonga, siSwati, Tshivenda, isiNdebele, English).
- **Department of Justice e-Filing Integration:** Secure digital submission of Form 2 Protection Order dossiers directly into Magistrates Court ICMS portals.

SCALABLE PUBLIC BENEFIT VISION

Strategic Horizon Outcome: The combination of immediate hackathon delivery (5 Core Features) and rigorous future epic architecture positions Umkhuseli as South Africa's most comprehensive, legally sound, and survivor-centric digital protection infrastructure.